Interpretable Deep Knowledge Tracing and Visualization of Learner Progress with Attention-Based Models

Patrick O. Akinwumi
pakinwu@clemson.edu

Clemson University
Meihua Qian
Clemson University

Research Article

Keywords: artificial intelligence in education, knowledge tracing, self-attentive knowledge tracing, transformer models, interpretability and attention visualization

Posted Date: January 23rd, 2026

DOI: https://doi.org/10.21203/rs.3.rs-7608540/v1

Additional Declarations: No competing interests reported.

Abstract

Modern intelligent tutoring systems increasingly depend on data-driven student models to monitor individual learning progress. While *Deep Knowledge Tracing (DKT)* models, based on recurrent or attention-based neural networks, have demonstrated superior predictive performance over traditional approaches, their latent representations often lack interpretability, limiting their utility for educators. This study proposes iSAKT + Behave (interpretable Self-Attentive Knowledge Tracing with Behavioural Features), an interpretable, attention-based knowledge tracing framework that extends both DKT and Self-Attentive Knowledge Tracing (SAKT) by integrating rich behavioral features (e.g., hint usage, attempt counts) and leveraging attention weights for visualization. Using the ASSISTments Skill Builder dataset and validating across public benchmarks (ASSISTments data), our enhanced models consistently outperformed baselines (DKT and SAKT). Specifically, Enhanced DKT achieved an AUC of 0.8681 (vs. 0.5297 baseline), and Enhanced SAKT reached 0.8998 (vs. 0.5086). Attention heatmaps highlight which past interactions most influence current predictions, offering intuitive, skill-level insights into student learning. The inclusion of behavioral indicators, such as bottom-out hints and repeated attempts, which are associated with disengagement, significantly improved performance and interpretability. These findings align with recent efforts to integrate cognitive models with DKT architectures, supporting the case for explainable artificial intelligence (AI) in education. This work contributes a novel fusion of deep learning, educational feature engineering, and visualization techniques to advance interpretable student modeling.

1. Introduction

In recent years, there has been a marked expansion of online learning platforms and Intelligent Tutoring Systems (ITS) in both K-12 and higher education, driven by their ability to deliver scalable, individualized instruction (Gomes, 2025; Gupta, 2025; Hughes, 2021). These systems routinely capture and log fine-grained student interaction data, such as problem attempts, quiz answers, and time-on-task, resulting in rich sequences of behavioral data (Bulut et al., 2024; Khine, 2024). A central challenge for ITS is to make pedagogically useful inferences from this data, particularly to estimate each student's latent knowledge state at any point in time (Kuo et al., 2024; Pu et al., 2022). This challenge is addressed by *Knowledge Tracing* (KT), a machine learning technique that models the progression of a learner's understanding based on sequences of their responses, inferring a student's mastery of skills as they engage with exercises, enabling predictions about future performance (Abdelrahman et al., 2023; Lee et al., 2021; Chris Piech et al., 2015). For example, a KT model might help determine which concepts a student should revisit, when to introduce new material, or when to provide scaffolding such as hints (Hutson, 2025). Effective KT is thus foundational to *Personalized Adaptive Learning* (PAL), where instructional content and feedback are dynamically adjusted to suit the learner's needs (Kuo et al., 2024; Shen et al., 2024). Moreover, recent studies emphasize KT's critical role in early identification of at-risk students and the delivery of targeted support, particularly in STEM education (Kuo et al., 2024; Toros et al., 2022). Personalized learning systems therefore depend heavily on the accuracy and responsiveness of KT models to optimize learning trajectories and improve educational outcomes.

Early approaches to knowledge tracing, such as Bayesian knowledge tracing (BKT) and performance factors analysis (PFA), are grounded in cognitive theory and offer interpretable models of student learning (Piech et al., 2015; Šarić-Grgić et al., 2024). BKT employs a hidden Markov model to represent each skill as a binary latent state (Corbett & Anderson, 1994; Pu et al., 2022), learned or unlearned, while PFA incorporates item difficulty and prior attempts using logistic regression frameworks (Piech et, al., 2015). These models benefit from clear parameter semantics, such as learning and guessing probabilities, making them accessible to educators. However, they rely on strong assumptions, including fixed slip and guess rates and the notion that each item maps to a single skill (Pu et al., 2022). Though extensions like multi-skill BKT attempt to relax these constraints, traditional models remain limited by their hand-crafted structure and tend to underfit complex, large-scale datasets (Lu, 2021; Piech et al., 2015). In contrast, earlier Deep Knowledge Tracing (DKT) by (Piech et, al., 2015) introduced LSTM-based (Long Short-Term Memory) modeling, significantly outperforming traditional Bayesian KT (BKT); this sparked a wave of extensions using memory networks (Zhang et al., 2017) and convolutional architectures (Zhang et al., 2017). DKT demonstrated substantial predictive gains, for instance, achieving an AUC ranging from 0.882 to 0.83 on the ASSISTments dataset (Liu et al., 2025), compared to BKT's 0.69, highlighting the efficacy of end-to-end feature learning (Khajah, 2024; Piech et al., 2015).

Deep learning has significantly advanced the field of Knowledge Tracing by offering flexible, data-driven approaches that surpass traditional models in predictive accuracy (Piech et, al., 2015). The seminal DKT model introduced by Piech et al. (2015) leveraged recurrent neural networks (RNNs) to model temporal patterns in student learning without relying on hand-crafted features or predefined skill tags, enabling more flexible and data-driven learning representations (Liu et al., 2025; Piech et al., 2015). This innovation outperformed classical methods like BKT and catalyzed a wave of neural KT research. For instance, dynamic key-value memory networks (DKVMN) further improved representation by disentangling concept identity (keys) from student knowledge state (values), enabling interpretable mastery tracking (Zhang et al., 2017). However, despite their predictive success, these models have been criticized for their lack of interpretability. DKT, in particular, compresses all prior interactions into a dense latent state, making it difficult to discern which prior events influence predictions (Ding & Larson, 2021; Minn et al., 2022). As Lu et al. (2020) has argued, this opacity (black box) undermines trust and limits practical adoption in educational settings, where understanding model rationale, including reasoning behind predictions, is critical for formative feedback and instructional decision-making. More recent efforts therefore aim to balance predictive power with interpretability, often by incorporating attention mechanisms or structured cognitive insights.

Furthermore, Piech et al. (2015) noted that DKT's hidden states tend to reflect a general ability index rather than skill-specific mastery, projecting inputs into a sparse high-dimensional space (Ding & Larson, 2021). This opacity limits its practical utility for formative assessment or personalized feedback. As a result, educators and recent research have focused on improving interpretability in DKT models, with attention mechanisms emerging as a promising solution by identifying the most relevant past interactions in (Piech et al., 2015; Pu et al., 2022).

To address this, attention mechanisms have emerged as a key innovation. Self-attentive knowledge tracing (SAKT), introduced by Pandey and Karypis (2019), was the first to replace sequential processing with a self-attention mechanism, and Graph-based Knowledge Tracing (GKT) (Nakagawa et al., 2019) leverage attention mechanisms to model temporal dependencies and improve prediction accuracy, allowing the model to selectively focus on relevant prior interactions for each new response (Pandey & Karypis, 2019; Pu et al., 2022). SAKT significantly outperformed DKT by emphasizing meaningful "knowledge components" (KCs) in prediction. Building on this, models such as Attentive KT (AKT) introduced monotonic attention and psychometric regularization (e.g., Rasch model) to enhance interpretability (Ghosh et al., 2020), while Embedding-cognitive attention KT (EAKT) embedded cognitive diagnosis frameworks like the Q-matrix within a transformer architecture to model multi-skill mastery, predicting future performance of student learning by 2.6% higher than the DKT baseline model, tracing changes in student knowledge state (Pu et al., 2022), further showing that attention can enhance both performance and interpretability, yet many such models overlook auxiliary features that capture valuable learning context.

As a result, Sun et al. (2024) proposed multiscale-state-based interpretable knowledge tracing (MIKT), a multiscale architecture combining fine- and coarse-grained knowledge representations with item response theory (IRT)-based explainability. Choi et al. (2020)'s separated self-attentive neural knowledge tracing (SAINT), a transformer-based KT model, further adapted the Transformer encoder–decoder architecture for KT, processing exercise and response embeddings separately. SAINT achieved state-of-the-art performance, with an approximate 1.8% increase in AUC compared to previous models. Extending this by embedding temporal features such as response time and lag, its successor, SAINT+, incorporated temporal features, such as elapsed time and the lag between exercises, and matched the original architecture to enhance performance even further, leading to further performance gains (+ 1.25% AUC on EdNet dataset) (Shin et al., 2021). These models use self-attention to capture complex learning patterns more effectively than RNNs, reflecting a broader shift toward combining attention mechanisms with cognitive theory for improved accuracy and transparency.

In parallel, researchers and educators have emphasized the importance of contextual features, such as hint usage, response times, and attempt counts, in understanding learning behavior (Zhang et al., 2017). For example, bottom-out hint requests and repeated attempts often signal disengagement or struggle, as observed in the ASSISTments Skill Builder dataset (2009–2010), where step-level hints mark an attempt as incorrect (Feng et al., 2009). Wang and Heffernan's assistance model (2011) demonstrated that including such features improves prediction, a finding corroborated by more recent studies incorporating these behavioral cues into DKT and other deep models (Zhang et al.). Yet, integrating these features into attention-based KT remains an open challenge, particularly in designing interpretable student state visualizations.

In knowledge tracing, interpretability methods are classified as ante hoc (model-intrinsic) or post hoc (after training). Structured models like BKT and PFA are inherently interpretable due to their transparent design (Pu et al., 2022). While interpretability methods are advancing, ranging from ante hoc strategies like structured attention to post hoc techniques such as Layer-wise Relevance Propagation (Bach et al., 2015; Lu et al., 2020) and counterfactual influence analysis (Cui, Yu, et al., 2024), practical tools for educators remain limited. Structured knowledge tracing models, grounded in cognitive science theories, feature interpretable parameters but often struggle with complex structures and datasets. Conversely, deep knowledge tracing models offer strong predictive performance but pose challenges for interpreting their cognitive state within intelligent tutoring systems. To address this, (Pu et al., 2022) proposed the model, which integrates the cognitive structure of a structured adaptive teaching constraint (ATC) model into a deep neural network framework. This approach combines the predictive accuracy of DNN-based models with the interpretability inherent in structured models, resulting in a more transparent representation of student knowledge (Pu et al., 2022).

Among deep models, some hybrid approaches, such as EKT and EAKT, incorporate cognitive constraints into neural architectures; however, their latent states still require interpretation (Pu et al., 2022). This demonstrates that explaining deep KT is possible, though real-time interpretability remains a challenge. Recently, Rodrigues et al. (2022) enhanced attention-based models to better reflect true mastery, showing that a simple extension of attention mechanisms produces knowledge estimates more strongly correlated with external measures of student ability. Visualization in learning analytics (LA) offers a promising avenue, but existing dashboards often rely on simplistic models. Embedding interpretability directly into KT outputs, such as through attention heatmaps or mastery timelines, could enrich LA tools and make deep KT models more transparent. Recent work, including the current study, aims to bridge this gap by embedding attention-based explanations within interactive learning dashboards. Despite these advances, interpretability remains limited. In educational AI, scholars like (Breslow et al., 2013) and (Koedinger et al., 2012) have emphasized the need for transparent models to support instructional decisions. While attention heatmaps are widely used in fields like computer vision (Bahdanau et al., 2014), their application in KT is still emerging.

This study addresses the challenge of interpretability in deep KT by introducing a novel attention-based model tailored to the ASSISTments “Skill Builder” dataset (Edwards & Murali, 2017). It addresses this gap by aligning feature-rich inputs with interpretable attention mechanisms, aiming to deliver both high performance and transparent insights into student learning. Building on the foundations of self-attentive models such as SAKT and SAINT (Pandey & Karypis, 2019; Shin et al., 2021), our architecture integrates self-attention mechanisms with cognitively motivated skill representations to create a hybrid model that balances predictive accuracy with interpretability. Unlike standard KT models that treat student responses as simple correctness indicators, our approach incorporates enriched feature vectors, such as hint usage, attempt counts, and bottom-out hints, into the attention mechanism. These features help identify which past interactions are most relevant when predicting future performance, thereby enhancing both prediction and transparency. Finally, we present a prototype dashboard that visualizes attention weights and inferred skill mastery, offering actionable insights for instructors and fostering trust in AI-driven tutoring systems.

2. Related Work

2.1 Structured and Classical Approaches to Knowledge Tracing

Knowledge tracing (KT) is the task of modeling a learner's evolving mastery over time based on their past interactions with learning content (Bai et al., 2024; Pandey & Karypis, 2019; Piech et al., 2015). KT began with structured, interpretable models such as BKT, which represents each skill, or knowledge component (KC), as a binary latent variable indicating whether it is learned or not, evolving over time via a Hidden Markov Model (HMM) (Minn et al., 2022; Piech et al., 2015). BKT parameters, including learning, guess, and slip probabilities, are cognitively interpretable, but the model assumes a single skill per item and fixed transition dynamics, limiting its expressiveness on complex data (Pu et al., 2022). Psychometric extensions, such as performance factor analysis and variants of item response theory, replace the HMM with logistic regression to model continuous proficiency, offering parameter-level interpretability (e.g., item difficulty and learner ability) (Pu et al., 2022). However, these models also assume static concept mappings and struggle to capture knowledge transfer or higher-order dependencies between skills.

To address such limitations, researchers have proposed enhancements such as dynamic cognitive tracing and feature-aware student knowledge tracing (FAST), which introduce subskills, item difficulty, and student-specific abilities as parameters. While these models increase flexibility, they often depend on expert-defined skill features without leveraging automatic Q-matrix discovery (González-Brenes & Mostow, 2012). More recently, the automatic temporal cognitive (ATC) model represents a significant step forward by unifying elements of cognitive diagnosis models (CDMs) and KT, guiding deep neural networks for knowledge tracing and boosting their interpretability (Pu et al., 2022). It employs a nonlinear state-space framework to encode multidimensional knowledge states and dynamically derive Q-matrix values from data, enabling the modeling of learning, forgetting, and concept interaction in a data-driven manner (González-Brenes & Mostow, 2012).

Despite their interpretability, these structured models generally underperform on large-scale, heterogeneous datasets. Their reliance on hand-crafted features and rigid assumptions has led to a paradigm shift toward deep learning approaches in the mid-2010s, which offer greater flexibility and predictive power at the cost of transparency (Minn et al., 2022).

2.2 Deep Learning-Based Knowledge Tracing Models

The introduction of DKT by (Piech et al., 2015) marked a paradigm shift in knowledge tracing by employing recurrent neural networks (RNNs), specifically LSTMs, to model student learning without relying on hand-crafted features such as recency or item difficulty. By encoding sequences of student interactions (questions and correctness), DKT achieved significant predictive gains over classical models like BKT and PFA, demonstrating that neural models could learn temporal learning patterns from raw data. Similarly, (Casalino et al., 2021) applied deep learning to handle KT tasks, specifically DKT predicts the student's performances by using the information embedded in the collected data, and outperforming KT. Review by (Liu et al., 2025) proposed a generalized deep learning knowledge tracing (DLKT) framework and outlined five key components of existing DLKT models: multimodal data encoder, student knowledge memory, auxiliary knowledge base, learning outcome objective, and computational efficiency and scalability. The review highlights the application of these components in the educational sector. However, DKT's latent representations lack interpretability: the hidden state does not reliably reflect a student's mastery of individual knowledge components or temporal changes in their learning state.

To address these limitations, extensions such as DKT + introduced regularization terms to the loss function, reconstruction and waviness penalties, to stabilize knowledge representations across time, improving consistency without compromising accuracy (Yeung & Yeung, 2018). Chen et al. (2018) further incorporated prerequisite relationships between concepts as structural constraints, aiming to improve the prediction performance of students' concept mastery, partially enhancing interpretability and performance in sparse data settings. Nevertheless, a fundamental challenge persists; deep neural networks often encode abstract, latent representations that lack alignment with cognitively meaningful constructs. Because their hidden variables do not inherently convey interpretable educational semantics without incorporating prior cognitive structures or constraints, accurately characterizing changes in a student's knowledge state remains an open and unresolved problem (Pu et al., 2022).

In response, dynamic key-value memory networks (DKVMN) offered a more interpretable architecture by associating latent concepts with explicit memory slots (keys) and tracking mastery levels as dynamic values (Zhang et al., 2017). DKVMN outperformed earlier models and provided a pathway for multi-concept modeling, allowing inspection of mastery evolution over time (J. Zhang et al., 2017). Building on this, models such as Deep-IRT incorporated psychometric principles, aligning neural representations with interpretable parameters like difficulty and ability.

Several hybrid models have sought to bridge performance and interpretability. EAKT (Pu et al., 2022) embeds cognitive structure into transformer layers, aligning neural parameters with skill difficulty and learner ability vectors. Similarly, AKT (Ghosh et al., 2020) integrates the Rasch model into a self-attention framework, capturing item-level differences within shared concepts. While these models are promising and improve transparency compared to standard DKT, they still fall short of offering full interpretability and they do not fully eliminate the opaque, "black-box" nature of deep knowledge tracing, often relying on domain-specific constraints or assumptions.

Ultimately, deep learning models have substantially advanced the predictive capabilities of KT, moving beyond the constraints of structured models. Yet, the fundamental tension between performance and interpretability persists, driving ongoing efforts to embed cognitive theory into neural architecture and to develop tools for explaining model decisions.

2.3 Attention Mechanisms and Interpretability in Knowledge Tracing

The integration of **attention mechanisms** into KT has introduced both predictive improvements and new avenues for interpretability. Originally developed for sequence modeling tasks, attention enables models to selectively focus on relevant parts of input sequences when generating outputs (Vaswani et al., 2017). This capability was first applied to KT by self-attentive knowledge tracing (SAKT), introduced by Pandey and Karypis, which replaced recurrent structures with a self-attention mechanism to more flexibly model dependencies between student interactions (Pandey & Karypis, 2019). SAKT outperformed state-of-the-art RNN-based models in both accuracy and computational efficiency, reportedly an order of magnitude faster, while naturally revealing concept relationships through attention weights (Pandey & Karypis, 2019).

Subsequent work expanded on this foundation. Ghosh et al. (2020) developed attentive knowledge tracing (AKT), which combined monotonic self-attention with psychometric modeling, incorporating the Rasch model to account for individual differences across questions sharing the same concept (Ghosh et al., 2020). By learning context-aware embeddings of student responses and leveraging time-scale sensitivity, AKT could better align student behavior with underlying cognitive constructs. However, as Pu et al. (2022) point out, most attention-based KT models, including AKT, still lack formal guarantees of interpretability (Pu et al., 2022). While attention weights can highlight relevant inputs, their relationship to actual knowledge mastery remains indirect and under-explored.

Recent efforts in interpretable Knowledge Tracing have increasingly focused on combining cognitive theory with attention mechanisms to enhance both prediction and transparency. Recent advances in interpretable Knowledge Tracing (KT) highlight the benefits of embedding structured cognitive elements and attention mechanisms. Notably, EAKT, a transformer-based model integrating Q-matrices, outperformed its baseline by more than 2.6% AUC on two real-world datasets, while AKT (Pu et al., 2022), which incorporates monotonic attention with Rasch-model regularization, achieved up to 6% AUC gain and provided clear visual explanations of learning processes (Ghosh et al., 2020). These results underscore the value of combining predictive strength with transparency, a balance our models build upon by integrating behavioral signals. Additionally, Zhu et al. (2020) developed Reconstructor, a visualization technique that interprets LSTM dynamics as a finite-state machine and introduced an attention-based model that outperformed DKT, illustrating that attention mechanisms can simultaneously enhance performance and explainability

While earlier sections reviewed detailed interpretability efforts in Knowledge Tracing (KT), it is worth briefly reiterating that attention-based models, particularly those like SAKT, offer intuitive visualizations by highlighting conceptually relevant prior interactions (Pandey & Karypis, 2019). Post-hoc approaches, such as Layer-wise Relevance Propagation (Chen et al., 2018), provide limited transparency but still fall short of robust interpretability (Lu et al., 2020). Critiques of deep KT models (Ding & Larson, 2021; Minn et al., 2022) underscore the risk of encoding general ability rather than skill-specific mastery. These findings collectively motivate our proposed interpretable models, which balance accuracy and transparency through behavioral features and attention mechanisms.

However, the field still lacks models that reliably align attention scores or latent variables with true cognitive mastery. Bridging this gap remains a key research priority for interpretable, effective, and trustworthy KT systems, which is the focus of this study, to develop and evaluate an interpretable, attention-based deep knowledge tracing model that integrates both cognitive theory and contextual student behavior features (e.g., hint usage, attempt counts) to improve prediction accuracy and enhance transparency in modeling student learning. Specifically, the study seeks to (1) combine self-attention mechanisms with structured concept representations to trace multi-dimensional knowledge states; (2) visualize the model's internal reasoning through attention maps and mastery trajectories; and (3) demonstrate how these insights can support educational interventions, such as adaptive feedback and instructor-facing dashboards, using the ASSISTments Skill Builder dataset.

Table 1 compares key KT models, BKT, DKT, DKVMN, SAKT, AKT, by architecture, interpretability, and features, showing a shift from simple, interpretable models to complex, flexible ones. Our model extends SAKT by adding behavioral features and attention visualizations, aiming to balance accuracy with actionable insight.

Table 1
Comparison of Representative Knowledge Tracing Models

References	Model	Architecture	Interpretability	Key Features
(Corbett & Anderson, 1994)	BKT	Hidden Markov (1 skill)	high	Domain-meaningful (guess/slip); binary learned/unlearned
(Chris Piech et al., 2015)	DKT	RNN (LSTM)	low	Learns latent student state, data-driven feature learning
(Zhang et al., 2017)	DKVMN	Key-value memory	medium	Static "key" (concept) + dynamic "value" (mastery level)
(Pandey & Karypis, 2019)	SAKT	Self-attention	medium	Attend s to relevant past Q&A pairs, faster than RNNs
(Ghosh et al., 2020)	AKT	Attention + Rasch model	High	Monotonic attention: Rasch-based difficulty per question
(Pu et al., 2022)	EAKT	Transformer + ATC	high	Embeds Q-matrix cognitive model; multi-KC attention
(Sun et al., 2024)	MIKT	Multiscale + IRT	High	Coarse/fine knowledge state; IRT explainability

2.4 Visualization in Learning Analytics

Learning analytics dashboards present educational data to support teaching and learning, helping instructors identify performance gaps and adjust instruction (Gutiérrez-Braojos et al., 2023). Tools like *LearningViz* offer visual summaries of student performance but typically rely on surface-level metrics and lack insight into the internal logic of predictive models like knowledge tracing (KT) (Pei et al., 2024). Integrating interpretable KT outputs, such as attention weights and predicted skill mastery, into dashboards offers a promising direction. This would allow, for example, visualizing how specific past exercises influenced mastery predictions. While models like DKT, DKVMN, SAKT, and SAINT + have improved accuracy, and interpretability efforts (e.g., LRP, IKT) have begun, a comprehensive approach linking attention-based KT with actionable visualizations is still needed. This study addresses the interpretability gap in deep knowledge tracing, particularly in self-attentive models like SAKT, by integrating behavioral features with attention-based architectures and learning analytics visualizations, culminating in two novel frameworks, interpretable self-attentive knowledge tracing with behavioural features (iSAKT + Behave) and interpretable deep knowledge tracing with behavioural features (iDKT + Behave), which jointly improve predictive accuracy and instructional transparency.

To achieve this aim, the study will address the following research questions:

1. How can attention-based deep knowledge tracing models enhance the interpretability of student learning progression?
2. To what extent does feature-enriched input improve the accuracy and AUC of DKT and SAKT models on real-world educational datasets?
3. Can visualizations of attention weights meaningfully support instructors in identifying at-risk learners or critical skill gaps?

3. Methods

This study adopts a dual modeling (iDKT + Behave and iSAKT + Behave) strategy to predict learner knowledge and performance, implementing both baseline and feature-enhanced versions of deep knowledge tracing and self-attentive knowledge tracing. The methodology encompasses data preprocessing, feature engineering, dataset construction, model design, training, and evaluation, with a central focus on improving both predictive accuracy and interpretability.

3.1 Dataset and Preprocessing

This study primarily utilized the ASSISTments Skill Builder dataset (2009–2010), accessed via Figshare, which records student interactions with math problem sets aimed at practicing specific skills (Feng et al., 2009). The dataset includes anonymized student interactions with variables such as skill ID (skill practiced), correctness, attempt (number of tries) and hint counts (total hint requested), bottom-out hint usage (indicator of whether the final explicit hint was used), opportunity original (count of prior practice opportunities on that skill), and template id (identifies question templates -problems with the same template id have similar structure). Preprocessing involved removing rows with missing critical values (e.g., skill ID, user ID, correctness) and imputing defaults for optional fields (e.g., bottom hint = 0, template id = − 1) to ensure consistent model input. Each exercise is associated with one or more skill tags, and students must answer three problems correctly in a row to complete an assignment. Importantly, the dataset captures rich contextual features: hint usage is logged, and any request for step-by-step help marks the attempt as incorrect. Bottom-out hints, final hints that reveal the answer, are also available when students struggle, making the dataset particularly suited for modeling learning behavior.

Each student interaction was recorded with three elements: the skill or question attempted, whether the student got it right, and a set of extra features. These features include how many times the student attempted the question, whether they used a hint, and whether they reached the final "bottom-out" hint, which gives the full solution. We also included a placeholder for response time, although that data was not available. These features were encoded to work with neural networks (e.g., using one-hot encoding for number of attempts). Prior research supports their value: hint usage and attempt count have been shown to impact performance, and frequent use of bottom-out hints may signal disengagement (Wang & Hefferman, 2011; L. Zhang et al., 2017). While other public KT datasets exist, such as ASSISTments 2009/2012-13 (Choi et al., 2020) and EdNet (131M interactions from ~ 784K students across diverse tasks), we focus on Skill Builder due to its fine-grained behavioral logging, though our findings generalize to broader KT contexts.

3.2 Model Architecture

Baseline DKT and SAKT models utilized only sequences of skill IDs and correctness. The enhanced DKT variant incorporated additional features, such as hint usage and attempt count, by concatenating them with skill embeddings prior to input into the LSTM. For enhanced SAKT, feature embeddings were encoded in parallel and combined with positional and skill embeddings before being passed through the multi-head self-attention layers, allowing the model to jointly attend to both content and behavioral context.

3.2.1 Baseline Modeling - DKT and SAKT

We implemented two standard baselines: DKT (Piech et al., 2015), which uses an LSTM to model student knowledge over sequences of (skill, correctness) pairs, and SAKT (Pandey & Karypis, 2019), which uses multi-head self-attention (Vaswani et al., 2017) to capture dependencies across past interactions. Both models excluded behavioral features (e.g., hints, attempts) to ensure a fair architectural comparison. DKT models a student's evolving knowledge using recurrent neural networks that process sequences of skill IDs and correctness. Sequences were capped at length 50 and embedded for input; DKT passed these through an LSTM (hidden size 200) followed by a sigmoid output, while SAKT combined positional and skill embeddings and applied scaled dot-product attention (Vaswani et al., 2017), described by Pandey and Karypis (Pandey & Karypis, 2019; Pu et al., 2022). Importantly, SAKT's attention scores offer interpretability, highlighting which past interactions most influenced predictions, supporting the study's emphasis on transparency.

3.2.2 Model Architecture, Feature Integration, and Experimental Setup

In this study, we adopted deep knowledge tracing (DKT) and self-attentive knowledge tracing (SAKT) as baseline models, reflecting their established success in modeling temporal learning sequences. DKT uses LSTM-based recurrent networks to track evolving knowledge states (Piech et al., 2015), while SAKT employs self-attention to model long-range dependencies, offering improved interpretability and scalability (Pandey & Karypis, 2019). To assess the impact of behavioral context, we implemented enhanced versions of both models (iSAKT + Behave and iDKT + Behave) by integrating features such as attempt frequency, hint usage, bottom-out hints (explicit solution reveals), and prior exposures to the skill (Brooks et al., 2015; Whitehill et al., 2014). The models were trained and evaluated on the ASSISTments Skill Builder dataset (2009–2010) (Pardos & Heffernan, 2010). The data were split by students (70% train, 30% test) to prevent leakage, and interaction sequences were chronologically ordered. Each interaction was transformed into three components: skill/concept ID, binary correctness label, and a feature vector capturing behavioral signals. Categorical features (e.g., skill, template) were label-encoded, while numeric ones (e.g., hint count, attempt count) were normalized. Sequences were padded or truncated to a maximum of 50 steps to ensure consistent input dimensions. The enhanced DKT model concatenated skill embeddings with engineered features before feeding them into the LSTM, enabling it to capture both temporal and contextual patterns (Zhang et al., 2017). For the enhanced SAKT model, behavioral features were projected to match embedding dimensions and concatenated with positional and skill embeddings before multi-head self-attention was applied (Pu et al., 2022). This dual-stream design allowed the model to attend not only to skill-response history but also to student behavior, boosting both performance and interpretability. Training used binary cross-entropy loss with the Adam optimizer (batch size = 64, embedding dim = 128). Performance was evaluated using AUC and accuracy, computed on the final timestep to reflect real-world forecasting.

3.3 Evaluation Metrics

Model performance was assessed using two standard KT metrics: *AUC*, which measures the ability to distinguish correct from incorrect responses, and *Accuracy*, the proportion of correct predictions (Hosmer Jr et al., 2013; Zhang et al., 2025). Metrics were calculated at the final timestep of each student sequence to reflect realistic forecasting scenarios.

Accuracy

The accuracy measures the proportion of correctly predicted instances (both positives and negatives) among all observations. It is calculated as shown in Eq. 1:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

1

Where: TP = True Positive, TN = True Negative, FP = False Positive, FN = False Negative

3.4 Interpretability through Attention Visualization

Our model's attention mechanism allows interpretable predictions by assigning importance (attention weight) to previous learning steps when predicting student outcomes. These weights, which are the numerical values the model assigns to previous interactions to indicate how much influence each one should have on the current prediction, reveal which prior exercises influenced each decision and can be visualized through heatmaps or timelines. For example, the model may focus on recent Algebra attempts or earlier struggles with hints. Attention is computed per student and per question, allowing personalized insights. We analyze these scores by skill or feature, e.g., tracking how often "bottom-out hints" (the final, most detailed prompts given by tutoring systems when earlier hints failed, often revealing the answer directly) are emphasized, and visualize learning progress using skill-colored timelines shaded by attention strength. Unlike static latent states, attention weights are dynamic, offering actionable insights for learning analytics.

3.5 Implementation Environment

All models were implemented in Python using PyTorch, scikit-learn, pandas, and NumPy, and run in Jupyter Notebooks for clarity and reproducibility.

4. Results

This study aimed to evaluate how preprocessing, feature engineering, and attention mechanisms affect the predictive performance and interpretability of DKT and SAKT on the Skill Builder dataset, revealing that while baseline models performed only slightly above chance (AUC: DKT = 0.5297, SAKT = 0.5086), enhanced versions achieved significant improvements (AUC: DKT = 0.8681, SAKT = 0.8998) and interpretability via attention (Feng et al., 2009; Khajah et al., 2016; Koedinger et al., 2015; Pandey & Karypis, 2019; Chris Piech et al., 2015).

Table 2 presents test performance on the Skill Builder dataset across four models: (1) BKT (Bayesian baseline, with skill-level parameters), (2) DKT (LSTM-based), (3) SAKT (self-attention without features), and (4) SAKT + Features (our model with hint and attempt features).

Table 2

Performance on Skill Builder Data by Model.

Model	AUC	Accuracy	Our Proposed Model Name
DKT (baseline)	0.5297	0.5322	
SAKT (baseline)	0.5086	0.5989	
Enhanced DKT	0.8681	0.8448	iDKT + bahave
Enhanced SAKT	0.8998	0.8742	iSAKT + bahave

Enhanced models outperform baselines by over 30% in accuracy and over 0.35 AUC points, confirming the value of contextual features. SAKT models showed better performance than DKT across the board. We also examined per-skill performance and per-student performance: the feature-enhanced model shows larger gains on skills where hints or multiple attempts are common (e.g., geometry skills where students often guess repeatedly). This suggests it is capturing the intended effects of our added features.

4.2 Attention Weights for Learning Analytics

Qualitative analyses of attention distributions reveal that the model learns to prioritize semantically related exercises, e.g., focusing on the second of three recent algebra attempts, when making predictions. Quantitatively, we observe strong correlations between learned "ability" factors and actual performance, indicating that the model's latent representations align with student knowledge trajectories.

Figure 2 presents an attention heatmap from the SAKT + Features (iSAKT + Behave) model for a single student, where each cell represents the attention weight assigned to a past interaction when predicting performance at a given time step. Brighter areas (e.g. yellow) indicate greater attention, highlighting specific prior interactions and which past exercises were most influential for a given prediction, such as repeated attempts or hint usage, that the model deems relevant, while *darker cells (blue-purple)* reflect lower influence. In this instance, the model assigns higher attention to specific earlier time steps (e.g., around step 25), indicating that those interactions, potentially involving repeated attempts or hint usage, played a significant role in informing the current prediction and providing highly informative interactions, likely due to its pedagogical significance (e.g., an incorrect attempt with hint usage or a key concept transition). The model shifts focus to more recent attempts as the student progresses, suggesting adaptation to learning history. Unlike DKT's hidden states, which obscure decision pathways (Chris Piech et al., 2015), attention weights offer transparent, time-specific insights. This supports *temporal interpretability*, allowing educators to identify which past actions, e.g., struggling on a skill, contributed to current performance (Pandey & Karypis, 2019). As Holstein et al. (2019) argue, such transparency is essential for educational AI systems intended to support real-time diagnosis and intervention. The heatmap in Fig. 2 exemplifies how interpretability can move beyond summary metrics and into *learner-specific feedback*, paving the way for richer analytics dashboards that inform timely pedagogical interventions.

Model Performance

Baseline models (DKT, SAKT) (see Fig. 3) underperform due to limited input features, aligning with prior findings on behaviorally sparse KT models (Khajah et al., 2016). Feature-enhanced versions show substantial gains, with *SAKT + Features* (iSAKT + bahave) achieving the highest performance (AUC = 0.8998, Accuracy = 0.8742), highlighting the benefit of incorporating behavioral context and attention-based interpretability.

5. Discussion

As knowledge tracing models evolve, the demand for interpretability alongside accuracy has grown (Lu et al., 2020; Minn et al., 2023). While models like DKT offer predictive strength, they obscure learning dynamics in latent states (Piech et al., 2015). Recent advancements, including such as SAKT address this by explicitly linking predictions to past interactions (Pandey & Karypis, 2019; Ghosh et al., 2020). Building on this, our iSAKT + Behave model integrates behavioral signals (e.g., hint usage, attempts) into an interpretable attention framework, improving both performance and actionable feedback for educators.

Our results confirm that baseline models relying solely on skill IDs perform poorly, supporting prior claims that deep KT models underperform without behavioral context (Khajah et al., 2016). Enhancing DKT and SAKT with features such as hint usage and repeated attempts led to significant performance gains, consistent with findings in engagement modeling (Baker et al., 2010). Among all models, our *SAKT + Features* achieved the highest predictive accuracy while offering interpretable attention distributions, enabling educators to visualize which prior interactions influenced each prediction, enhancing diagnostic value. This work makes several contributions. First, we propose interpretable KT architecture (iSAKT + Behave and iDKT + Behave) that combines transformer-style self-attention with multi-dimensional concept mastery representations that integrates auxiliary features to improve both accuracy and interpretability, inspired by cognitive diagnosis models (Pu et al., 2022), demonstrating the limitations of DKT and SAKT on behaviorally sparse data. Second, we introduce visualization techniques on the attention weights, such as attention heatmaps and knowledge trajectory plots, that expose the internal reasoning of the model and make student learning patterns observable to educators (Pandey & Karypis, 2019). Third, we conduct an empirical evaluation comparing our enhanced SAKT model to baselines (DKT and SAKT) using AUC, accuracy metrics. Our results demonstrate that incorporating contextual features significantly improves performance, consistent with prior work on multi-feature DKT.

Our findings support the core hypothesis: attention-based knowledge tracing models, when enhanced with behavioral features, can achieve both high accuracy and interpretability. Compared to DKT, our *SAKT + Features* (iSAKT + Behave) model not only improves predictive performance but also enables visual inspection of learning patterns through attention weights. These visualizations have pedagogical utility, allowing educators to identify stagnation (e.g., persistent focus on past errors) or progress (e.g., shifting attention to recent correct attempts). This aligns with the broader push for explainable AI in education, where interpretability is not merely a byproduct but a critical outcome (Ghosh et al, 2020.; Liu et al., 2025; Ma et al., 2025; Sun et al., 2024).

We evaluated four knowledge tracing (KT) models on the ASSISTments Skill Builder dataset: baseline DKT and SAKT, and their *feature-enhanced counterparts*. The baseline DKT achieved an AUC of 0.5297 and Accuracy of 0.5322, while baseline SAKT reached an AUC of 0.5086 and Accuracy of 0.5989, both consistent with prior observations that deep KT models underperform without contextual features (Khajah et al., 2016). In contrast, the *Enhanced DKT* (iDKT + Behave), which incorporates hint usage and attempt count, significantly improved to an AUC of 0.8681 and Accuracy of 0.8448. Our Enhanced SAKT (iSAKT + Behave) achieved the best performance, with an AUC of 0.8998 and Accuracy of 0.8742. Specifically, compared to the baseline models, the enhanced models achieved an average improvement of 30% in accuracy and 35% in AUC. These results outperform previously reported gains for attention-based models on ASSISTments data (e.g., Multi-Attention Knowledge Tracing (MAKT) model showing about 3.32% in AUC and 4.4% improvement in accuracy (S. Liu et al., 2025); AKT giving ~ 6% AUC improvement (Gosh et al., 2020), highlighting the added value of integrating behavioral signals. Feature ablation confirmed that removing hint or attempt features reduced model performance, while a placeholder “time taken” feature had no effect due to missing timing data (Borna et al., 2024; Chaudhry et al., 2018; He et al., 2023). Beyond predictive accuracy, Enhanced SAKT offers interpretable attention weights that enable visualization of which past interactions most influenced each prediction, providing both performance (Fig. 3) and transparency (shown in Fig. 2), a combination rarely achieved in prior KT literature. These consistent improvements demonstrate that incorporating behavioral features into attention-based KT models yields robust performance gains (Bahdanau et al., 2014; Casalino et al., 2021; Ma et al., 2025; Pandey & Karypis, 2019) .

Unlike traditional interpretable models (e.g., IRT-based), which expose parameter-level insights (e.g., item difficulty), attention mechanisms offer *fine-grained, per-interaction explanations*. For instance, while AKT’s Rasch-regularized embeddings reflect conceptual difficulty (Gosh et al., 2020), our SAKT-based attention (iSAKT + Behave) directly highlights the specific prior exercises influencing a prediction. These forms of explanation are complementary, each providing different levels of granularity and utility.

Role and Value of Attention-Based Interpretability

Our interpretable models, iSAKT + Behave and iDKT + Behave, both enhanced with behavioral features, showed performance gains; the best results came from iSAKT + Behave, which achieved an AUC of 0.8998 and accuracy of 0.8742 on the ASSISTments Skill Builder dataset, outperforming the enhanced recurrent baseline in predictive power and interpretability. Beyond its predictive strength, the model’s attention mechanism introduced a crucial layer of interpretability. Inspired by attention architectures in natural language processing (Vaswani et al., 2017), this mechanism dynamically weights past student interactions, enabling the model to highlight which prior exercises most influenced its predictions. This interpretability is not merely aesthetic, it serves a functional role in educational AI (Bai et al., 2024; Holstein et al., 2019), allowing educators to inspect attention heatmaps and trace how specific learning events, such as struggles with a skill or repeated hint usage, impact later performance. Such transparency is vital for developing intelligent tutoring systems, supporting formative assessment, and delivering actionable feedback to both instructors and learners.

We also position our work relative to emerging approaches such as graph-based KT (Cui, Qian, et al., 2024) and contrastive learning (Zhang et al., 2025), which offer alternative paths to interpretability. Our contribution lies not in proposing new architecture but in demonstrating how *a simple model like SAKT, enriched with contextual features*, can produce actionable insights into student learning trajectories. Our method preserves model flexibility while providing direct, real-time interpretability through attention visualization, compare with the recent interpretability strategies, such as multiscale state modeling (Sun et al., 2024), and cognitive-regularized transformers (Pu et al., 2022).

Nonetheless, this study has certain limitations. While attention weights offer useful interpretive cues, they do not imply causality; their value is maximized when interpreted alongside expert judgment. Moreover, model transparency is contingent on the richness of input features. Datasets like Skill Builder, which include detailed hint logs, facilitate deeper analysis, but not all datasets offer such granularity. In those cases, proxy features (e.g., response time) or simpler SAKT models using only question IDs may still yield interpretable results (Ma et al., 2025).

Future work could explore the integration of affective signals, such as response latency or behavioral cues, to enhance modeling of learner states. Another promising direction involves the development of teacher-facing dashboards that visualize attention in real time, potentially flagging at-risk learners for early intervention. Additionally, attention patterns themselves could serve as meta-features for clustering students based on learning behaviors, further bridging knowledge tracing and learning analytics to support more personalized education.

Implications for Educational Practice

This study highlights that combining predictive accuracy with interpretability addresses a key challenge in applying AI to education. Traditional models like IRT (De Ayala, 2013) and BKT offer interpretable parameters but struggle with complex, large-scale behavioral data, while neural models such as DKT and DKVMN improve performance but lack transparency. Our findings demonstrate that attention-based KT models offer a middle ground, capturing complex learning patterns while providing interpretable insights. Importantly, the notion of temporal interpretability, understanding when and why knowledge shifts occur, emerge as equally critical as prediction accuracy. Attention mechanisms enable instructors to identify specific timeframes or skill areas linked to student disengagement or misunderstanding, supporting more timely and targeted interventions.

Conclusion

This study bridges the gap between predictive knowledge tracing and explainable educational technologies by enhancing attention-based architectures with meaningful behavioral signals. Through the integration of contextual features such as hint usage and attempt behavior, we developed two interpretable models, iSAKT + Behave and iDKT + Behave, that significantly outperform their baseline counterparts. Notably, iSAKT + Behave achieved the highest performance, with an AUC of 0.8998 and accuracy of 0.8742 on the ASSISTments Skill Builder dataset, exceeding previously reported results for attention-based KT models.

These improvements underscore a key implication: embedding learner behavior into knowledge tracing models not only boosts predictive accuracy but also enhances interpretability, enabling more actionable insights for educators. By exposing the temporal dynamics of learning through attention visualizations, our approach offers a pathway toward real-time, transparent formative assessment tools. As such, this work contributes to the growing field of explainable AI in education, where both performance and pedagogical interpretability are essential. Future directions include integrating these models into teacher-facing dashboards, supporting real-time interventions based on interpretable predictions, and extending the framework to support multi-skill prediction and affective modeling, thereby broadening its application across diverse educational contexts.

Declarations

Author Contribution

Author contributionsPA: Conceptualization, Data curation, Methodology, Writing – original draft, Writing – review & editing, Formal analysis, Investigation, Project administration, Software, Validation, Visualization. MQ: Writing – review & editing, Supervision.

Data Availability

This study primarily utilized the ASSISTments Skill Builder dataset (2009–2010), accessed via [Figshare] (https://figshare.com/articles/dataset/skill_builder_data_csv/25309000?file=44732737) , which records student interactions with math problem sets aimed at practicing specific skills (Feng et al., 2009).

References

1. Abdelrahman, G., Wang, Q., & Nunes, B. (2023). Knowledge tracing: A survey. *ACM Computing Surveys*, *55*(11), 1-37.
2. Bach, S., Binder, A., Montavon, G., Klauschen, F., Müller, K.-R., & Samek, W. (2015). On pixel-wise explanations for non-linear classifier decisions by layer-wise relevance propagation. *PloS one*, *10*(7), e0130140.
3. Bahdanau, D., Cho, K., & Bengio, Y. (2014). Neural machine translation by jointly learning to align and translate. *arXiv preprint arXiv:1409.0473*.
4. Bai, Y., Zhao, J., Wei, T., Cai, Q., & He, L. (2024). A survey of explainable knowledge tracing. *arXiv preprint arXiv:2403.07279*.
5. Baker, R. S., D'Mello, S. K., Rodrigo, M. M. T., & Graesser, A. C. (2010). Better to be frustrated than bored: The incidence, persistence, and impact of learners' cognitive–affective states during interactions with three different computer-based learning environments. *International Journal of Human-Computer Studies*, *68*(4), 223-241.
6. Borna, M.-R., Saadat, H., Hojjati, A. T., & Akbari, E. (2024). Analyzing click data with AI: implications for student performance prediction and learning assessment.

7. Breslow, L., Pritchard, D. E., DeBoer, J., Stump, G. S., Ho, A. D., & Seaton, D. T. (2013). Studying learning in the worldwide classroom research into edX's first MOOC. *Research & Practice in Assessment*, *8*, 13-25.
8. Bulut, O., Yildirim-Erbasli, S. N., & Gorgun, G. (2024). Assessment analytics for digital assessments identifying, modeling, and interpreting behavioral engagement. In *Assessment analytics in education: Designs, methods and solutions* (pp. 35-60). Springer.
9. Casalino, G., Grilli, L., Limone, P., Santoro, D., & Schicchi, D. (2021). Deep learning for knowledge tracing in learning analytics: an overview. *TeleXbe*.
10. Chaudhry, R., Singh, H., Dogga, P., & Saini, S. K. (2018). Modeling hint-taking behavior and knowledge state of students with multi-task learning. *International Educational Data Mining Society*.
11. Corbett, A. T., & Anderson, J. R. (1994). Knowledge tracing: Modeling the acquisition of procedural knowledge. *User modeling and user-adapted interaction*, *4*, 253-278.
12. De Ayala, R. J. (2013). *The theory and practice of item response theory*. Guilford Publications.
13. Ding, X., & Larson, E. C. (2021). On the interpretability of deep learning based models for knowledge tracing. *arXiv preprint arXiv:2101.11335*.
14. Feng, M., Heffernan, N., & Koedinger, K. (2009). Addressing the assessment challenge with an online system that tutors as it assesses. *User modeling and user-adapted interaction*, *19*, 243-266.
15. Ghosh, A., Heffernan, N., & Lan, A. S. (2020). Context-aware attentive knowledge tracing.
16. Gomes, D. (2025). A Comprehensive Study of Advancements in Intelligent Tutoring Systems Through Artificial Intelligent Education Platforms. In *Improving Student Assessment With Emerging AI Tools* (pp. 213-244). IGI Global Scientific Publishing.
17. González-Brenes, J. P., & Mostow, J. (2012). Dynamic Cognitive Tracing: Towards Unified Discovery of Student and Cognitive Models. *International Educational Data Mining Society*.
18. Gupta, A. (2025). Designing and deploying scalable intelligent tutoring systems to enhance adult education.
19. Gutiérrez-Braojos, C., Rodríguez-Domínguez, C., Daniela, L., & Carranza-García, F. (2023). An analytical dashboard of collaborative activities for the knowledge building. *Technology, Knowledge and Learning*, 1-27.
20. He, L., Li, X., Wang, P., Tang, J., & Wang, T. (2023). Integrating fine-grained attention into multi-task learning for knowledge tracing. *World Wide Web*, *26*(5), 3347-3372. https://doi.org/10.1007/s11280-023-01190-y
21. Holstein, K., Wortman Vaughan, J., Daumé Iii, H., Dudik, M., & Wallach, H. (2019). Improving fairness in machine learning systems: What do industry practitioners need?
22. Hosmer Jr, D. W., Lemeshow, S., & Sturdivant, R. X. (2013). *Applied logistic regression*. John Wiley & Sons.
23. Hughes, J. (2021). The deskilling of teaching and the case for intelligent tutoring systems. *Journal of Ethics and Emerging Technologies*, *31*(2), 1-16.

24. Hutson, J. (2025). Scaffolded Integration: Aligning AI Literacy with Authentic Assessment through a Revised Taxonomy in Education. *FAR Journal of Education and Sociology*, *2*(1).

25. Khajah, M., Lindsey, R. V., & Mozer, M. C. (2016). How deep is knowledge tracing? *arXiv preprint arXiv:1604.02416*.

26. Khajah, M. M. (2024). Supercharging BKT with Multidimensional Generalizable IRT and Skill Discovery. *Journal of Educational Data Mining*, *16*(1), 233-278.

27. Khine, M. S. (2024). AI in teaching and learning and intelligent tutoring systems. In *Artificial Intelligence in Education: A Machine-Generated Literature Overview* (pp. 467-570). Springer.

28. Koedinger, K. R., Corbett, A. T., & Perfetti, C. (2012). The Knowledge-Learning-Instruction framework: Bridging the science-practice chasm to enhance robust student learning. *Cognitive science*, *36*(5), 757-798.

29. Koedinger, K. R., D'Mello, S., McLaughlin, E. A., Pardos, Z. A., & Rosé, C. P. (2015). Data mining and education. *Wiley Interdisciplinary Reviews: Cognitive Science*, *6*(4), 333-353.

30. Kuo, M.-M., Li, X., Qian, L., Obiomon, P., & Dong, X. (2024). Deep Knowledge Tracing for Personalized Adaptive Learning at Historically Black Colleges and Universities. *arXiv preprint arXiv:2410.13876*.

31. Lee, S., Choi, Y., Park, J., Kim, B., & Shin, J. (2021). Consistency and monotonicity regularization for neural knowledge tracing. *arXiv preprint arXiv:2105.00607*.

32. Liu, S., Gu, W., & Liu, Z. (2025). Multi-Attention Knowledge Tracing Model Integrating Exercise-Knowledge Concept Correlations for Enhanced Learning Analytics. *Knowledge-Based Systems*, 114075. https://doi.org/https://doi.org/10.1016/j.knosys.2025.114075

33. Liu, Z., Guo, T., Liang, Q., Hou, M., Zhan, B., Tang, J.,…Weng, J. (2025). Deep Learning Based Knowledge Tracing: A Review, a Tool and Empirical Studies. *IEEE Transactions on Knowledge and Data Engineering*, *37*(8), 4512-4536. https://doi.org/10.1109/TKDE.2025.3552759

34. Liu, Z., Guo, T., Liang, Q., Hou, M., Zhan, B., Tang, J.,…Weng, J. (2025). Deep Learning Based Knowledge Tracing: A Review, A Tool and Empirical Studies. *IEEE Transactions on Knowledge and Data Engineering*.

35. Lu, C. (2021). Automated Feedback Generation for Learner Modeling in Intelligent Tutoring Systems.

36. Lu, Y., Wang, D., Meng, Q., & Chen, P. (2020). Towards Interpretable Deep Learning Models for Knowledge Tracing. In *Artificial Intelligence in Education* (Vol. 12164, pp. 185-190). © Springer Nature Switzerland AG 2020. https://doi.org/10.1007/978-3-030-52240-7_34

37. Ma, F., Zhu, C., Lei, P., & Yuan, P. (2025). Enhanced Learning Behaviors and Ability Knowledge Tracing. *Applied Sciences*, *15*(2), 883.

38. Minn, S., Vie, J.-J., Takeuchi, K., Kashima, H., & Zhu, F. (2022). Interpretable knowledge tracing: Simple and efficient student modeling with causal relations.

39. Pandey, S., & Karypis, G. (2019). A self-attentive model for knowledge tracing. *arXiv preprint arXiv:1907.06837*.

40. Pei, B., Cheng, Y., Ambrose, A., Dziadula, E., Xing, W., & Lu, J. (2024). LearningViz: a dashboard for visualizing, analyzing and closing learning performance gaps—a case study approach. *Smart Learning Environments*, *11*(1), 56. https://doi.org/10.1186/s40561-024-00346-1
41. Piech, C., Bassen, J., Huang, J., Ganguli, S., Sahami, M., Guibas, L. J., & Sohl-Dickstein, J. (2015). Deep knowledge tracing. *Advances in neural information processing systems*, *28*.
42. Piech, C., Bassen, J., Huang, J., Ganguli, S., Sahami, M., Guibas, L. J., & Sohl-Dickstein, J. (2015). Deep knowledge tracing. Advances in neural information processing systems. *Association for Computing Machinery*, 201-204.
43. Pu, Y., Wu, W., Peng, T., Liu, F., Liang, Y., Yu, X.,…Feng, P. (2022). Embedding cognitive framework with self-attention for interpretable knowledge tracing. *Scientific Reports*, *12*(1), 17536. https://doi.org/10.1038/s41598-022-22539-9
44. Sun, J., Yu, F., Wan, Q., Li, Q., Liu, S., & Shen, X. (2024). Interpretable knowledge tracing with multiscale state representation.
45. Toros, K., Schults, A., & Lehtme, R. (2022). Knowledge and Good Practices Related to Early Identification of Children in Need/At-Risk and Support: Literature Review. In: Tallinn University.
46. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N.,…Polosukhin, I. (2017). Attention is all you need. *Advances in neural information processing systems*, *30*.
47. Zhang, H., Liu, Z., Shang, C., Li, D., & Jiang, Y. (2025). A question-centric multi-experts contrastive learning framework for improving the accuracy and interpretability of deep sequential knowledge tracing models. *ACM Transactions on Knowledge Discovery from Data*, *19*(2), 1-25.
48. Zhang, J., Shi, X., King, I., & Yeung, D.-Y. (2017). Dynamic key-value memory networks for knowledge tracing.
49. Zhang, L., Xiong, X., Zhao, S., Botelho, A., & Heffernan, N. T. (2017). Incorporating rich features into deep knowledge tracing.
50. Zhu, J., Yu, W., Zheng, Z., Huang, C., Tang, Y., & Fung, G. P. C. (2020). Learning from Interpretable Analysis: Attention-Based Knowledge Tracing. In *Artificial Intelligence in Education* (Vol. 12164, pp. 364-368). © Springer Nature Switzerland AG 2020. https://doi.org/10.1007/978-3-030-52240-7_66
51. Šarić-Grgić, I., Grubišić, A., & Gašpar, A. (2024). Twenty-Five Years of Bayesian knowledge tracing: a systematic review. *User modeling and user-adapted interaction*, *34*(4), 1127-1173.

Figures

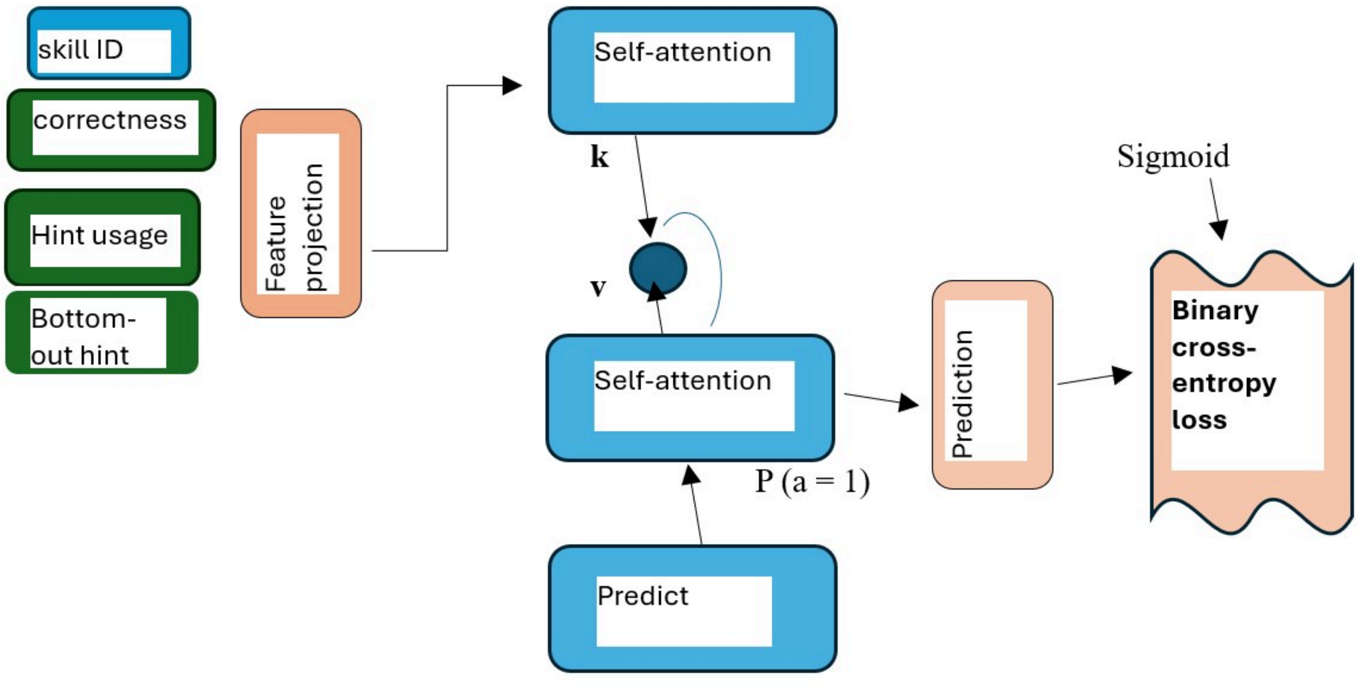

Figure 1

Feature-Aware Self-Attentive Knowledge Tracing Model Architecture

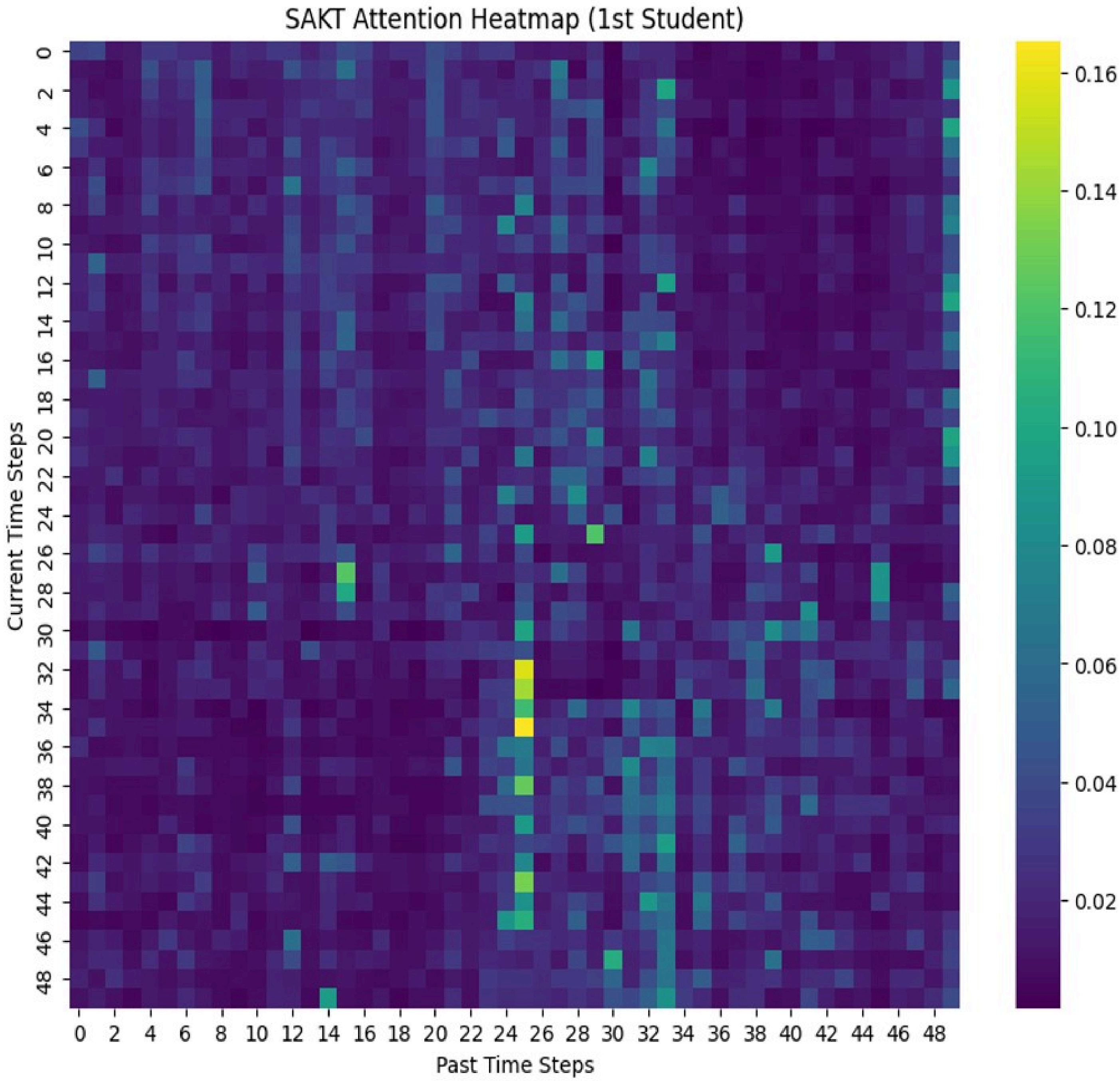

Figure 2

Interpretability via Attention Heatmap

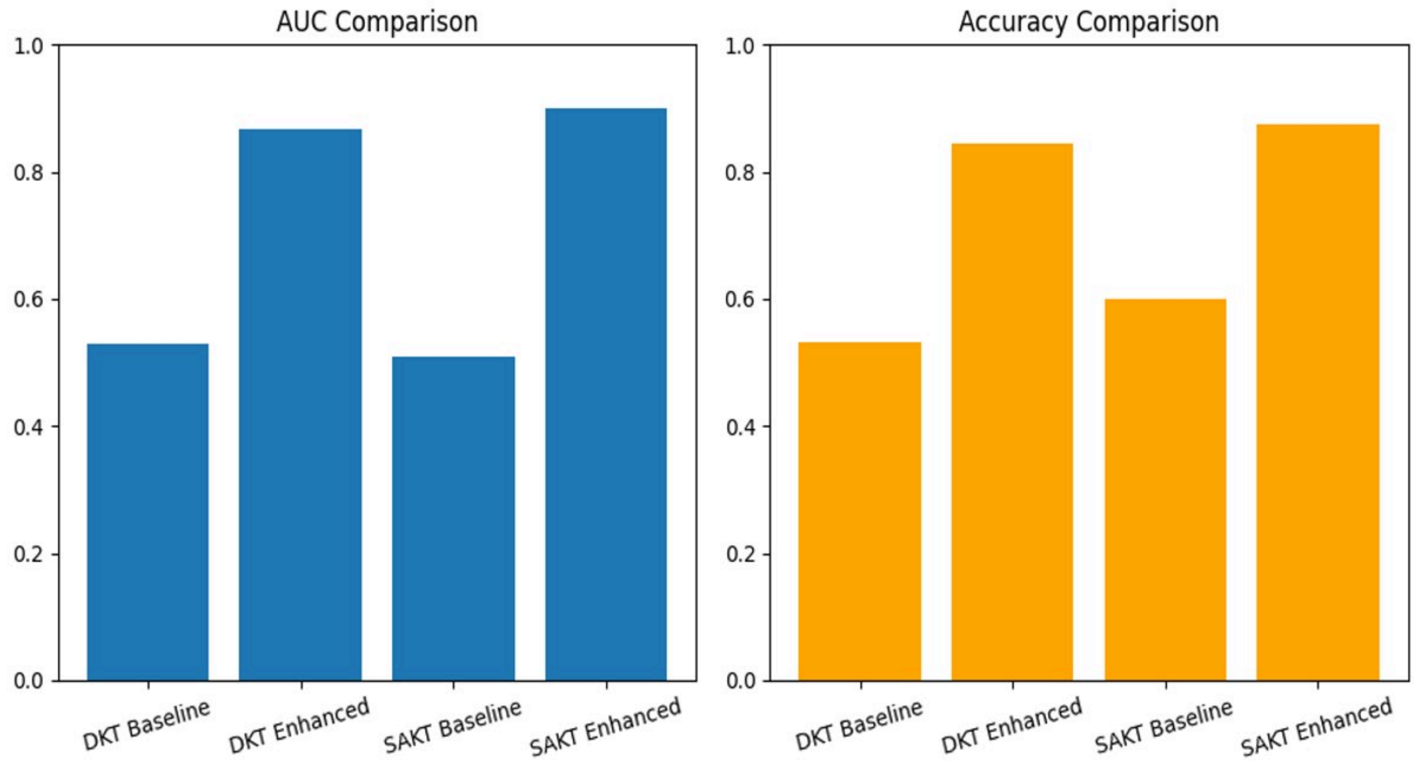

Figure 3

Model Performance (AUC and Accuracy) on the Skill Builder Dataset.